

Self-Study DA2: Evaluation of Programs for the Unemployed

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Active labour market programs are widely used to improve employment prospects of unemployed, yet estimating their causal effects can be challenging. In this self-study project, you will analyze a hypothetical training program implemented in three regions (A, B, C), each with potentially different assignment and participation mechanisms. In a pilot region, training was offered randomly and participation was voluntary, while in the other regions assignment was determined by caseworkers and participation was mandatory.

The objective of this project is to understand which identification strategies are credible in different settings and how treatment effect estimates change when assumptions are strengthened or violated.

The dataset for this self-study is available on Canvas. Please use the dataset assigned to you. Submit your Python code (main file and functions file) together with a written report in PDF format. The report should not exceed 3 pages in Part I and 3 pages in Part II (tables and figures do not count toward the page limit). The report must clearly present results, interpretation, and a concise explanation of your data preparation and analysis steps.

You are expected to write your own functions for estimating treatment effects, standard errors, test statistics, and for producing plots. For intermediate procedures (e.g., propensity score estimation, matching), standard packages may be used. You may reuse functions from the PC Labs.

Data Description

The dataset contains individual-level observations from a labour market training program. Key variables include:

<code>region</code>	Region A, B, or C
<code>offer</code>	=1 if randomly offered training
<code>takeup</code>	=1 if individual participated in training
<code>outcome</code>	Labour market outcome after five years
<code>age</code>	Age in years
<code>gender</code>	Gender indicator
<code>baseline_income</code>	Pre-program income
<code>unemployment_duration</code>	Duration of previous unemployment

The raw data contains several other covariates, which might be important as well.

Part I: Selection on Observables, Due: 23.03.2026

1. Data Preparation

Prepare the data to be suited for the analysis. This involves filtering for **missing values**, handling of **categorical data**, correct miscoded or **inconsistent variables**, checking for **outliers**, removing **post-treatment variables**, etc. Clearly document your decisions and explain (if applicable) how they affect identification and interpretation.

2. Exploratory Data Analysis

Provide a comprehensive descriptive analysis:

- Compare covariate distributions across Regions A, B, and C.
- Present summary statistics and visualizations.
- Examine overlap and support both graphically and numerically.

Is there a region that causes problems? What's going on?

3. Estimation under Selection on Observables

Using the regions with well behaved propensity scores, estimate treatment effects using the following methods:

- Raw mean differences,
- Regression adjustment,
- Inverse probability weighting (IPW),
- Doubly robust estimation.

For each method report estimated effects and standard errors and p values. Discuss the identifying assumptions required for each estimator and compare the estimated quantities.

4. Credibility of the CIA

Think carefully about whether the selection-on-observables assumption might hold in this setting. You might consider (This is not about testing/coding but about a thorough discussion, relating to findings in Q2):

- Whether there are ways to assess the CIA in a data driven way.
- How similar the treated and control groups appear on the observed characteristics and how that would help.
- What aspects of the program or its implementation could make the assumption more or less plausible.
- Whether some parts of the data seem more consistent with the assumption than others.

5. Alternative Identification Strategies

Given your prior evaluation, which alternative strategy would be suitable to get credible causal effects in this data set (or subgroups of it)? Discuss an alternative strategy.